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Some Studies on Energy Savings in Sponge Iron Plants

This paper presents the potential for energy savings in various capacity sponge iron plants by estimating heat transfer rates using stabilized and optimized temperature data within the rotary kiln/reactor. Air jet seal instead of mechanical seal arrangements at specific locations in the rotary kiln and cooler are proposed and this is shown to reduce the power consumption and the size of the centralized grease lubricating system. The paper proposes atomized water spray on the post combustion chamber instead of larger droplets to reduce the size of the water handling system, thereby saving energy. The paper also suggests optimization of the performance of a pneumatic coal injector by optimizing coal and air quantities. [DOI: 10.1115/1.1577601]

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Introduction

One of the avenues for increasing the productivity of the iron and steel sector lies in establishment of the rotary kiln based DR plant for sponge iron making, requiring less costly and abundantly available noncoking coal. Hence, the establishment of mini steel plants using a noncoking coal based DR plant is now gaining importance. Although most of these plants have acquired the desired level of operational efficiency, the energy utilization of various units is found below of the optimum limit. Hence, an energy audit is attempted in this study to optimize the energy utilization in some of the important units of established noncoking coal based DR plants and also to estimate the unit wise energy consumption and saving opportunities. The study presented in this paper attempts to estimate the scope of energy savings in some areas, in and around the rotary kiln, as this is the main energy consuming part of a noncoking coal based sponge iron plant. This is done by estimating (i) the optimum fuel/reductant, combustion air rate for the process requirement in rotary kiln, (ii) the size of Induced Draft (ID) fan for exhaust gas treatment, (iii) air requirement for the coal injection and combustion for utilization of unburnt fuel at the post combustion chamber (PCC). Energy losses are further minimized by (a) estimating the frictional loss along the mechanical sealing surface around the inlet and outlet end of the rotary kiln and cooler and incorporation of an air jet in some of these areas, (b) optimizing the performance of coal injector and (c) using atomized spray cooling along the PCC instead of droplet size water spray cooling.

System Description

Figure 1 illustrates the operating principle of the rotary kiln. The incoming bed material consisting of ore (6–25 mm) and a major portion of feed coal (12–20 mm) is heated not only by heat transfer through radiation and convection from the flue gas and wall, but also due to heat generated by the combustion of coal within the bed of material [1–4]. Nearly half the portion (lengthwise) of the kiln is considered as the preheated zone. In this zone, the incoming bed of iron ore is heated to about 1000°C. The remaining part of unburnt feed coal within the bed of material and some fresh amount of coal (about 9 mm size in plants with capacity $\geqslant$100 TPD and 3 mm size in smaller plants with capacity $<$100 TPD) is injected pneumatically over the heated bed along the outlet portion of the kiln that serves to reduce the heated ore [5,6]. The process of such transformation from iron oxide/ore to iron is endothermic. Thus in order to sustain the temperature of the bed material at about 1000°C, the necessary air for combustion is admitted in the kiln's overboard space at a controlled rate from its shell mounted air fan [5,7]. Normally, there are 6 to 7 fans in the smaller plants and 10 to 12 in the larger plants. The air tube delivers air along the kiln axis towards either the outlet or inlet direction. The reduced product is finally discharged from the reactor into an indirectly cooled rotary cooler where it is cooled to about 70–90°C and is then fed into the product handling system.

Process exhaust gases from the kiln are subsequently ducted through the PCC where it is initially mixed with some fresh air and burnt with the help of a burning flame that is maintained by a set of oil/gas burners. Then, the hot flue gas is cooled using water spray and ducted through a scrubber in order to remove the particulate material as sludge. The clean flue gas is now ducted through ID fan and finally released to the atmosphere through a stack.

The reduced sponge iron is at about 200°C and is highly susceptible to oxidation, therefore any ingress of air cannot be allowed. The ingress of air is possible through the sealing areas between the intermediate stationary housings and the outlet end of rotary kiln and of the inlet end of the rotary cooler. Therefore, mechanical sealing (Fig. 2) is usually provided in the mentioned areas.

Scope of Energy Savings in Rotary Kiln

Some areas of the rotary kiln where energy savings are possible are analyzed and discussed in this section.

Heat is generated due to combustion of fuel within the kiln as follows:

- Combustion of solid coal ($C+O_2$ volatile matter$+O_2 \rightarrow CO_2$, H_2O, CO etc.) within and on the surface of the bed of material. Most of the coal combustion should take place in the preheating zone.
- Combustion of unburned flue gas ($CO+O_2$ (mainly) $\rightarrow CO_2$, $C_nH_m+O_2 \rightarrow nCO_2+\frac{1}{2}mH_2O$) along the overboard area of the kiln.
- Combustion or gasification of coal particles ($C \rightarrow CO$, C_nH_m

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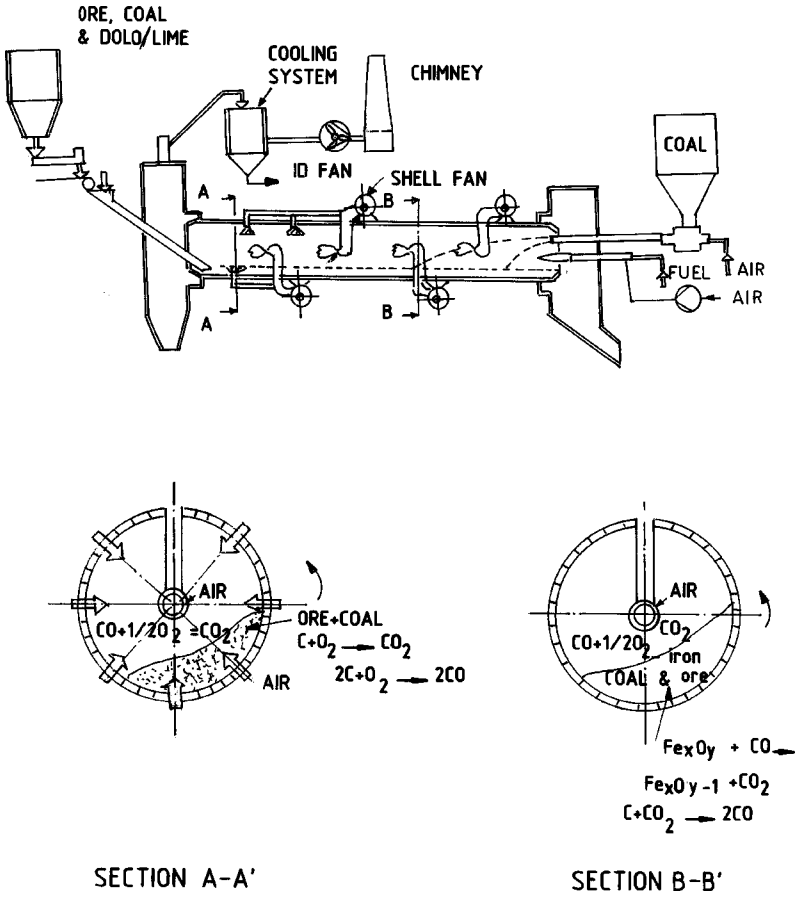


Fig. 1 Operating principles of process concerning noncoking coal based direct reduction in a rotary kiln

$\rightarrow nCO + m_1H_2 + 1/2\ m_2H_2O$) that are injected and sprayed over the heated bed of material, at the outlet end region of the reactor.

Heat, thus generated due to combustion of solid coal and volatile matters is transferred to the various components within the kiln and finally to the bed of material at relatively lower temperature. The mode of this heat transfer is very complex, as it not only involves the gas and solid but also the rotary motion of the kiln.

The total heat requirement of the process depends on (i) the heat generated due to combustion, (ii) heat gained by the incom-

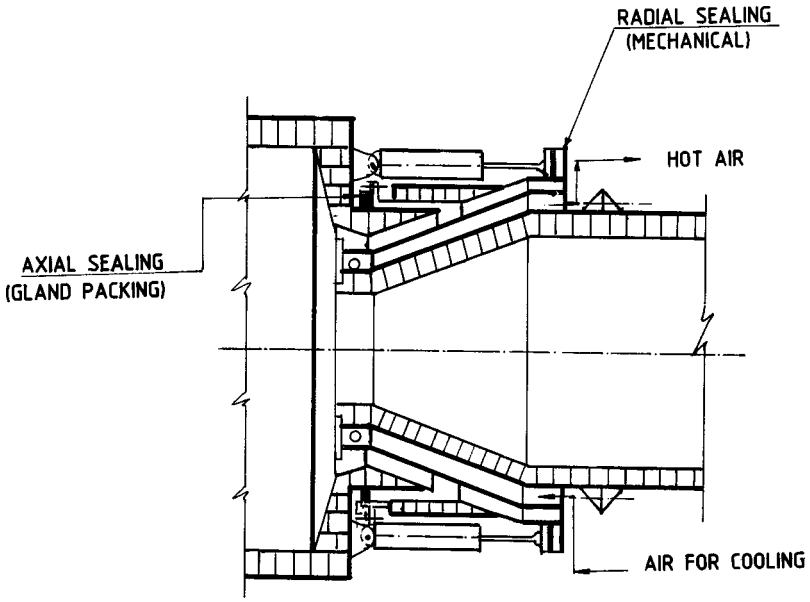


Fig. 2 The kiln seal arrangement

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Table 1 Measured inside temperature °C (±10°C variation) at ten equidistant locations in the kilns under stabilized operating conditions

	Kiln Size D(m), L(m) Plant Capacity Tonne per Day (TPD)	t_{g1}, t_{w1}, t_{s1}	t_{g2}, t_{w2}, t_{s2}	t_{g3}, t_{w3}, t_{s3}	t_{g4}, t_{w4}, t_{s4}	t_{g5}, t_{w5}, t_{s5}	t_{g6}, t_{w6}, t_{s6}	t_{g7}, t_{w7}, t_{s7}	t_{g8}, t_{w8}, t_{s8}	t_{g9}, t_{w9}, t_{s10}	t_{g10}, t_{w10}, t_{s10}
1	0.9, 15.0	980	1000	1020	1020	1040	1060	1080	1100	1100	1100
	5–8	580	680	790	880	940	1000	1040	1040	1050	1060
		370	500	630	760	870	940	990	1000	1000	1000
2	1.2, 22.0	950	990	1010	1010	1030	1050	1070	1090	1100	1100
	10–12	560	670	750	860	940	990	1020	1030	1050	1050
		350	490	620	750	870	940	980	990	1000	1000
3	2.6, 40.0	930	950	980	1020	1060	1070	1100	1100	1130	1130
	100	550	660	730	820	930	980	1020	1040	1060	1060
		300	450	580	730	840	910	980	990	1000	1000
4	3.75, 72.0	920	960	1000	1040	1100	1120	1140	1150	1150	1150
	300	470	640	750	830	940	980	1010	1060	1070	1070
		250	430	570	720	830	940	990	1000	1000	1000
5	4.25, 80.0	900	950	1000	1040	1080	1120	1140	1150	1170	1170
	400	460	630	740	820	940	1000	1020	1060	1080	1080
		250	430	570	710	820	940	990	1000	1010	1010

Where t is temperature in °C and subscript g, w, s indicate gas, wall and solid respectively.

ing air and bed of materials, (iii) heat lost through the reactor wall, (iv) heat carried away by the outgoing flue gas and (v) heat required for the endothermic reaction process of reduction to continue. The coal and air consumption rate is decided by the constituent energy balance requirement of the process. Combustion of flue gas overbed/overboard at selected locations and fuel within the bed of material increases the temperature of flue gas and bed respectively. Therefore, to estimate fuel and air consumption rate, plant data during stabilized operation at several equidistant locations (for example say ten) along the length of the reactor for various plants are required. These data for various plants and kiln sizes are obtained from Indian sponge iron industries and presented in Table 1. These data are taken as a basis for further study.

Due to variation in the temperatures of flue gas, wall and bed of material, the bed of material gains heat and attain the desired temperature in different locations of the kiln [7–9].

Estimation of Total Heat Requirement at any Stage

The estimation of total heat requirement consists of the following steps.

Heat Gained by the Bed of Material [10,11].

$$Q_s = M_s C_s (t_{sm} - t_{s\alpha}) \tag{1}$$

Where, Q_s is heat transfer rate (kcal/h) to solid, M_s is solid material feed rate (kg/h), C_s is specific heat of the solid, t_{sm} is the maximum temperature (°C) of solid inside the reactor and $t_{s\alpha}$ is the ambient temperature (°C) of the solid.

Heat Lost Through the Kiln Wall and Refractory [11].

$$Q_{lk} = (t_{ki} - t_{ko}) \Bigg/ \left[\left(\frac{1}{h_{ri} A_{ri}} \right) + \left(\frac{1}{2\pi K_r L} \right) \ln \left(\frac{D_{si}}{D_{ri}} \right) + \left(\frac{1}{2\pi K_s L} \right) \ln \left(\frac{D_{so}}{D_{si}} \right) + \left(\frac{1}{h_\alpha A_{so}} \right) \right] \tag{2}$$

Where, Q_{lk} is rate of heat loss (kcal/h) through kiln/reactor wall, t_{ki} and t_{ko} are inside and outside temperatures (°C) of the kiln shell respectively, h_{ri} and h_α are heat transfer coefficients (W/m^2°C) of rotary kiln at inside refractory and at the outer shell respectively. A_{ri} and A_{so} are areas (m^2) of rotary kiln at inside the refractory and at the outer shell respectively, L is length (m) of the rotary kiln, K_r and K_s are thermal conductivities (W/m°C) of refractory and kiln shell respectively, D_{si} and D_{ri} are inside diameter (m) of kiln shell and inside diameter (m) of the refractory respectively. D_{so} is outside diameter (m) of the shell. h in Eq. (2) is given by

$$h = N_u . K/D, \quad N_u = C[G_r . P_r]^n \tag{3}$$

Where, h is heat transfer coefficient (W/m°C), N_u is Nusselt number, K is thermal conductivity (W/m°C), D is diameter (m), G_r is Grashof number, P_r is Prandtl number and $C = 0.54/0.25$, $n = 0.13/0.33$ for Laminar/Turbulent range [11].

The actual heat lost through the wall includes the heat lost through kiln shell, inlet and outlet hood, and PCC and inlet area of the cooler. Therefore, the total heat loss is assumed as twice that of the kiln in the study and hence Q_{lt} is taken as two times the Q_{lk}.

Heat Required for the Reduction Process to Continue [12,13]. The stoichiometric equations responsible for reduction of iron ore are:

$$1/2\ Fe_2O_3 + 3/2\ CO = Fe + 3/2\ CO_2 - 12{,}866\ \text{kJ/kmol} \tag{4}$$

$$1/3\ Fe_3O_4 + 4/3\ CO = Fe + 4/3\ CO_2 - 3{,}904\ \text{kJ/kmol} \tag{5}$$

$$FeO + CO = Fe + CO_2 - 16{,}108\ \text{kJ/kmol} \tag{6}$$

$$FeO + C = Fe + CO + 156{,}482\ \text{kJ/kmol}\quad (37{,}336\ \text{kcal/kmol}) \tag{7}$$

Reaction (7) indicates the quantum of heat required for the desired rate of reduction of iron oxide to iron to take place. CO thus generated by the reaction (7) is utilized either for further reduction of ore vide reaction Eqs. (4) to (6) or for combustion ($2CO + O_2 \rightarrow 2CO_2$) and subsequent heating of overbed of material. It is revealed from the reaction (7) that 56 kg of sponge iron requires approximately 12 kg of carbon and 37,336 kcal of endothermic heat. Since noncoking coal contains approximately 70% of fixed carbon, the coal requirement is about 20 kg against production of 56 kg of sponge iron. Therefore, the endothermic process heat (kcal) required is:

$$Q_p = (37336/56) M_s \tag{8}$$

where, Q_p is rate of heat (kcal/h) required for the process to take place and M_s is the material feed rate (kg/h).

The coal required (kg) for reduction is $M_c = (20/56) M_s$

Where, M_c is material feed rate (kg/h) of coal.

$$\text{Heat carried away by flue gas, } Q_f = M_a C_a (t_{gp} - t_\alpha) \tag{9}$$

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Table 2 Estimated heat requirement [11] against heating of bed of material, heat loss from wall and heat of reaction etc. based on measured values of temperatures in Table 1 (Assumptions: $P_r=0.7$, $\rho_a=0.2947/0.9980$ kg/m^3 at 1000°C/100°C)

	D(m), L(m), Capacity (TPD)	Q_s kcal/day Eq. (1) (I)	Q_{lt} kcal/day Eq. (2) (II)	Q_p kcal/day Eq. (8) (III)	Q_t kcal/day (IV=I+II+III)
1	0.9, 15.0, 5–8	30×10^5	9×10^5	54×10^5	93×10^5
2	1.2, 22.0, 10–12	57×10^5	41×10^5	80×10^5	178×10^5
3	2.6, 40.0, 100	565×10^5	145×10^5	670×10^5	1380×10^5
4	3.75, 72.0, 300	1690×10^5	350×10^5	2000×10^5	4040×10^5
5	4.25, 80.0, 400	2260×10^5	425×10^5	2670×10^5	5350×10^5

Where, M_a is a air feed rate (kg/h), C_a is specific heat (kcal/kg°C) of air, t_{gp} and t_α are flue gas temperatures (°C) at PCC inlet and ambient respectively.

Total heat required Q_t for the process at any stage:

$$Q_t = Q_s + Q_{lt} + Q_p \tag{10}$$

Assuming the approximate calorific value of the coal as 6000 kcal/kg. combustion efficiency as 0.7 and the heat carried away by flue gas $Q_t = M_a C_a (t_{gp} - t_\alpha)$

The final empirical relation for estimating coal requirement is:

$$Q_t + M_a C_a (t_{g,p} - t_\alpha) + M_c C_c (1000 - t_\alpha) = M_c 6000(0.7) \tag{11}$$

Moreover, the minimum air that is required per kg of coal burnt [13,14] is $M_a = 11.6M_c + 34.8\,[H_2 - (O_2/8)] + 4.355$

The analysis of Indian noncoking coal used for sponge iron making has revealed that it contains approximately 70% carbon; 6–8% hydrogen and 7–8% oxygen. Therefore, the minimum air required for burning of this coal is about 11.1 kg/kg of coal [3,9]. Operational data of noncoking coal based plant has revealed that the process requires 1600 kg of iron ore and 800 kg of coal to produce a tonne of sponge iron. As air contains 21% oxygen and 1.6 kg ore releases 0.48 kg of oxygen, the actual air requirement for combustion of 1 kg of coal in DR process is about 7.75 kg. Hence, Eq. (11) may be modified as follows:

$$Q_t + M_a C_a (t_{g,p} - t_\alpha) + (M_a/7.75) C_c (1000 - t) = (M_a/7.75) 6000(0.7) \tag{12}$$

Using the methodology described above, the values of Q_s, Q_{lt}, Q_p and Q_t are estimated for various size plants (of Table 1) and presented in Table 2.

Mechanism of Heat Transfer. An understanding of the heat transfer mechanism inside a rotary kiln allows us to accurately estimate Q_s. The mechanism of heat transfer [7] is very complex because it not only involves the gas and solid, but also the wall of a rotating kiln. Apart from the conduction through kiln wall, the entire heat transfer involves five following components (Fig. 3) as follows:

- Gas-to-solid by radiation and convection (Q_{gs})
- Gas-to-wall by radiation and convection (Q_{gw})
- Exposed wall-to-solid by radiation (Q_{wes})
- Covered wall-to-solid by radiation and convection (Q_{wes})
- Surface-to-interior of the bed by bulk mixing through conduction and convection (Q_c)

Estimation of above mentioned heat transfer ($Q_{gs}, Q_{gw}, Q_{wes}, Q_{wes}$) is using the following equations.

(a) Gas-to-solid

The equation governing the heat transfer from gas to solid is [7,11]

$$\frac{Q_{gs}}{A_{se}} = \frac{\sigma}{\left(\frac{1}{\varepsilon_g} + \frac{1}{\varepsilon_s} - 1\right)} (\hat{T}_g^4 - \hat{T}_s^4) + h_{gs}(\overline{T}_g - \overline{T}_s) \tag{13}$$

Where, A_{se} is the exposed solid surface, σ is Stefan-Boltzman constant (kW/m^{2}K^4), ε_g and ε_s are emissivities of gas and solid respectively. Solid and gas emissivities are taken as 0.5 and 0.16 respectively [7]. $\hat{T}_g$ and $\hat{T}_s$ are the fourth power average temperature (K), $\overline{T}_g$ and $\overline{T}_s$ are the arithmetic average temperatures of gas and solid respectively, and subscript gs is used for gas to solid, se for solid exposed, g for gas and s for solid. h_{gs} is convective heat transfer coefficient (W/m^2°C) from gas to solid.

Also,

$$h_{gs} = (N_u K_g / D) \tag{14}$$

$$N_u = 1.86(R_e P_r)^{1/3} \quad \text{for } Re < 2000$$

$$N_u = 0.023\, R_e^{0.8} P_r^{0.4} \quad \text{for } Re > 2000 \text{ and} \tag{15}$$

$$\overline{T}_g = \frac{\sum_{i=1}^{n} T_{gi} A_i}{\sum_{i=1}^{n} A_i}; \quad \hat{T}_g^4 = \frac{\sum_{i=1}^{n} T_{gi}^4 A_i}{\sum_{i=1}^{n} A_i} \tag{16}$$

(b) Gas-to-wall

The equation governing the heat transfer from gas to wall is

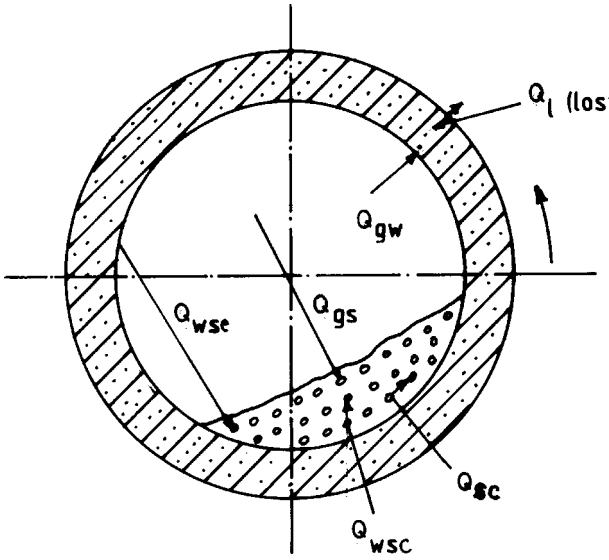


Fig. 3 Heat flow path in a rotary kiln

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Table 3 Estimated Plant wise value of various heat transfer rate (Eqs. (13) to (20)) based on temperature data presented in Table 1

	TDP/ & A_w, A_{se}, A_{sc}, A_c (m^2) approximate	Q_{gw} kcal/day Eq. (17) (I)	Q_{gs} kcal/day Eq. (13) (II)	Q_{ws} kcal/day Eq. (18, 19) (III)	Q_{cs} kcal/day (IV=I−III)	Q_s kcal/day (IV+III+II=V)	Q_s kcal/day Table 2 (VI)	Error % V−VI/VIX 100
1	5–8/30.0, 11.0,13.0, 24.0	23×10^5	12×10^5	15×10^5	8×10^5	35×10^5	30×10^5	14.2%
2	10–12 58.0, 21.0, 25.0, 46.0	48×10^5	22×10^5	26×10^5	16×10^5	64×10^5	57×10^5	12.2%
3	100/230.0, 85.0, 97.0,182.0	410×10^5	195×10^5	225×10^5	185×10^5	605×10^5	565×10^5	7.0%
4	300596.0,229.0, 252.0,481.0	1270×10^5	575×10^5	800×10^5	470×10^5	1845×10^5	1690×10^5	9.0%
5	400840.0, 300.0,350.0, 650.0	1745×10^5	785×10^5	1090×10^5	665×10^5	2545×10^5	2260×10^5	12.0%

$$\frac{Q_{gw}}{A_{we}} = \frac{\sigma}{\left(\frac{1}{\varepsilon_g}+\frac{1}{\varepsilon_w}-1\right)}(\hat{T}_g^4-\hat{T}_w^4)+h_{gw}(\overline{T}_g-\overline{T}_w) \qquad (17)$$

Where, A_{we} is the exposed wall surface, subscript gw is gas to wall and we is wall exposed. The wall emissivity may be assumed as 0.5 [7].

(c) Exposed wall-to-solid

The equation governing the heat transfer from exposed wall to solid is [7]

$$\frac{Q_{wes}}{A_{se}} = \frac{\sigma}{\left(\frac{1}{\varepsilon_w}+\frac{1}{\varepsilon_s}-1\right)}(\hat{T}_{wes}^4-\hat{T}_s^4) \qquad (18)$$

where, $\hat{T}_{wes}$ is temperature (K) of exposed surface of the kiln wall and subscript wes is exposed wall to solid, se is solid exposed, w is wall and s is solid.

(d) Covered wall-to-solid

The equation governing the heat transfer from covered wall to solid is

$$\frac{Q_{wcs}}{A_{sc}} = \frac{\sigma}{\left(\frac{1}{\varepsilon_w}+\frac{1}{\varepsilon_s}-1\right)}(\hat{T}_{wcs}^4-\hat{T}_s^4) \qquad (19)$$

Where, wcs is covered wall to solid and T_{wcs} is temperature (K) of covered surface of the kiln wall and $Q_{ws}=Q_{wcs}+Q_{wes}$, taking $T_w \approx T_{we}$, A_{se} is unexposed/covered solid surface, A_{sc} is covered solid surface and $A_s=A_{se}+A_{sc}$

(e) Surface-to-interior of the bed by bulk mixing

Heat transfer coefficient of the solid increases due to tumbling, rolling, slumping and sliding motion. Under these conditions, a heat balance on the surface particle give rise to the following equation:

$$\sigma S_s(\hat{T}_g^4-\hat{T}_s^4)A_s = V_s\rho_s C_s(dT_s/dt) \qquad (20)$$

Where, S_s is the shape factor of solid/gray body. V_s is the volume of solid (m^3), ρ_s is the bulk density (kg/m^3) of the solid and C_s is the specific heat of the solid (kcal/kg°C), dT_s/dt is the rate of increase of the temperature of the solid (K/h). Plant wise estimation of the above mentioned heat transfers is carried out and presented in Table 3.

Proposed Air Injection Distribution System

Q_s is the heat transfer required for the solid and the estimated value as mentioned in Table 3 is found to be slightly higher than that of Table 2. This indicates that higher amount of heat transfer is required than the estimated values given in Table 2. Combustion air distribution and the air requirement can be estimated using Eq. (12). The distribution of air is very important as it controls the temperature within the kiln. As mentioned earlier, coal used for the sponge iron making normally has a volatile matter content in the range 28–32%, large amount of these volatiles evolve along the inlet portion of the kiln i.e. in the preheating zone and some of these volatiles escape as waste gas without being fully utilized in the rotary reactor, which decreases the plant energy utilization capacity and thus increases overall coal consumption. In order to accomplish kiln operation under controllable temperature, most of the kilns are provided with an air distribution system. Table 4 summarizes plant wise estimated coal and air consumption rates. The process of combustion is controlled using an air fan as shown in Fig. 1. It is proposed that, a part of the required process air, say 20–30% is injected inside the bed of material through specially designed radial nozzles, which will cause the carbon as well as volatile components released from coal within the bed of material at temperature range 300°C to 600°C to burn directly. This mode of injection of air into the bed promotes combustion and heat generation within the bed of material.

A major portion of the remaining air of the order of 50 to 60% is introduced through the air tubes distributed uniformly along the axis of the kiln to burn most of the unburned flue gas (CO and part of volatile) in the free/overboard space and in the axial zone of the reactor. The remaining portion of the process air of about 10 to 30% is injected radially through the nozzles in the kiln free board space above its axial location. Thus, a major portion of the remaining unburnt carbon monoxide and volatile burn near the top

Table 4 Plant wise estimated coal and air consumption rate

	Plant Capacity TPD	Total Heat requirement kcal/kg (Min.) Table 2	Coal Consumption rate for reduction Eq. (8) TPD	Coal consumption rate for heating Eq. (12) TPD	Total coal requirement rate TPD	Air requirement m^3/day Eq. (12)	Max. no. of blower to be used (plant data)	Capacity of the suitable shell blower m^3/h
1	5–8	93×10^5	2.9	5.1	8.0	35×10^3	7	210–250
2	10–12	178×10^5	4.2	6.3	10.5	42.6×10^3	7	260–300
3	100	1380×10^5	36	49	85.0	31.4×10^3	8	1650–1750
4	300	4040×10^5	110	120	230.0	90×10^4	10	3750–4000
5	400	5350×10^5	150	155	305.0	120×10^4	10	5000–5225

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Table 5 Proposed radial injection facility along the preheating zone

TPD	No. of module (Radial air injection)	Total no. of blowers in preheated zone	Total no. of shell air blowers along the kiln	No. of shell air tube (axial) towards outlet end (Fig. 4)	No. of shell air tube (axial) towards inlet end (Fig. 4)
5–8	Only with 2nd shell air blower from the inlet end	3/4	6/8	1st and 2nd from the inlet end	3rd to 6/8th from the inlet end
10–15	Only with 2nd & 3rd shell air blowers from the inlet end	4	8	1st and 2nd from the inlet end	3rd to 8th from the inlet end
100	Only with 1st, 2nd & 3rd shell air blowers from the inlet end	4	8	1st to 3rd from the inlet end	4th to 8th from the inlet end
300	With 1st to 4th shell air blowers from the inlet end	5	10	1st to 4th d from the inlet end	5th to 10th from the inlet end
400	With 1st to 4th shell air blowers from the inlet end	5/6	10/12	1st to 4th from the inlet end	5th to 10/12th from the inlet end

region of the kiln i.e. near the refractory wall generates more heat. This proposed air injection distribution system for various plant capacities is summarized in Table 5.

It is envisaged that some of the fuel value of the flue gas may be utilized by increasing its retention time in the kiln. This may be achieved by injecting axial air through the shell blower towards the outlet end, instead of towards the inlet end, in the preheating zone. Therefore, the air distribution system as shown in Fig. 4 and presented in Table 5 is proposed. It is preferable to deliver air axially through only the shell air tube in the reducing zone towards inlet end as any radial injection in the reducing zone may re-oxidize the already reduced product at nascent state.

Pneumatic Coal Injection

The performance of a rotary kiln for DR to a considerable extent depends on the capability of the pneumatic coal injector (Figs. 1 and 4). Pneumatic injection of coal is achieved by utilizing kinetic energy/flow momentum of either more amount of air at low velocity or less amount of air at high velocity. Coal particles are sprayed over the heated bed of material mainly of iron ore in order to maintain reducing atmosphere. The power required for coal injection [6] can be estimated as follows:

Initial coal particle velocity:

$$v_{cp} = L/2[g_c/(D/2 - \text{bed depth})]^{1/2} \quad (21)$$

Where, g_c is acceleration due to gravity (m/sec^2), D is the kiln inside diameter (m) and L/2 is the length of the preheating zone.

Conveying/transportation velocity of conveying medium and coal particle is:

$$v_{cp} = v_{ac}[1 - 1.41(d_c/0.3038)^{0.3}(\rho_c/1000)^{0.5}] \quad (22)$$

Where, v_{ac} is the stabilized conveying velocity (m/sec) of air, d_c is the average diameter (mm) of the particles and ρ_c is its density (kg/m^3) and v_{cp} is the velocity of the coal particle.

Velocity of air injection through the nozzle of the injector using energy balance:

$$\frac{1}{2}M_a v_{an}^2 = \frac{1}{2}M_a v_{ac}^2 + \frac{1}{2}M_c v_{cp}^2 \quad (23)$$

Where, M_a and M_c are mass flow rate of air and coal respectively, v_{an} and v_{ac} are the velocities of air at nozzle and conveying duct and $v_{an} = v_{ac}^2 + (M_c/M_a)v_{cp}^2$

The streams of coal particles leave the outlet of the injector in the free board space of the intermediate housing (coupled to the kiln outlet end) as a free turbulent jet. The longitudinal velocity of air free jet at a distance X from injector exit may be estimated as:

$$\frac{v_{acx}}{v_{aci}} = 2.28\left[\frac{d_i}{X}\right]^{0.5} \quad \text{for } 5 > X/d_i < 2000 \quad (24)$$

where V_{acx} is the velocity of air at a distance X, V_{aci} is the velocity at injector exit and d_i is the injector diameter.

The coal particles are subjected to a drag force depending on the surrounding air. The drag force is given by

$$F_d = c_d\frac{\rho_a V_{cpr}^2}{2g_c} \quad (25)$$

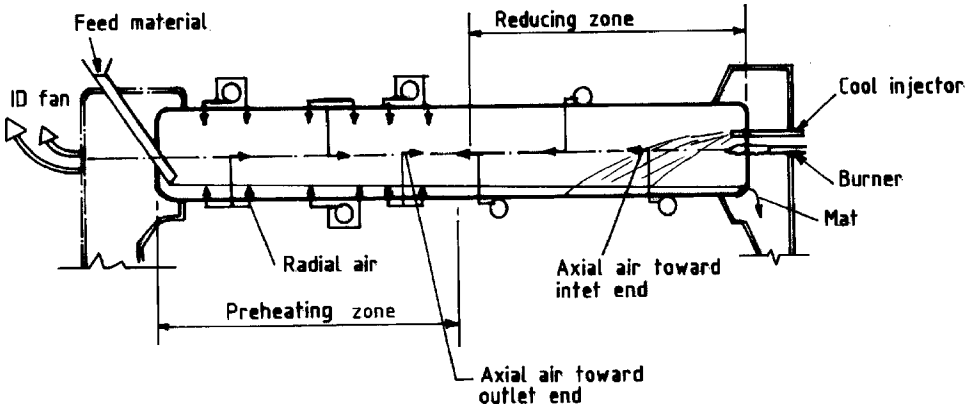


Fig. 4 Schematic of proposed air distribution system

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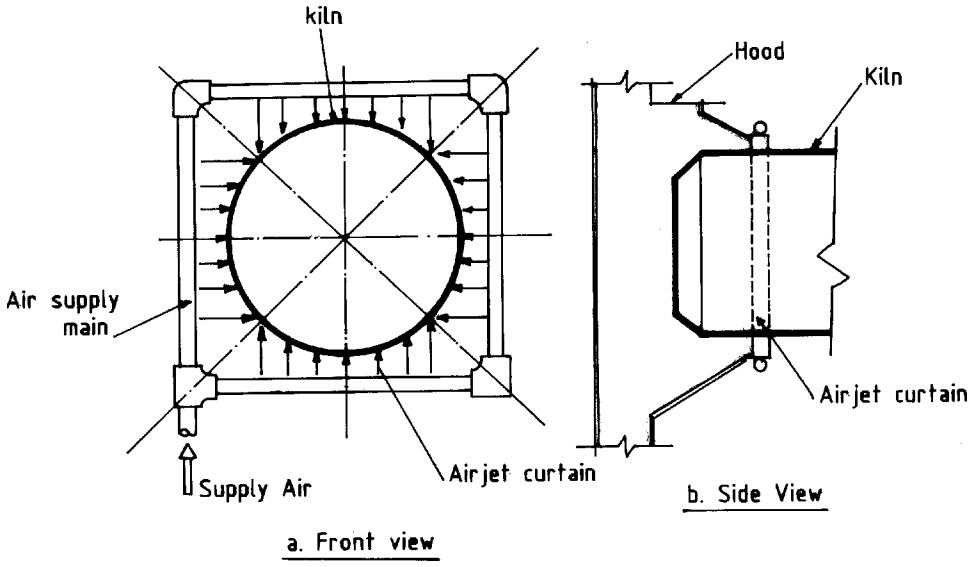


Fig. 5 Schematic of air jet sealing

Where c_d is the coefficient of drag, ρ_a is the density of air and v_{cpr} is the relative velocity of coal particle with respect to the surrounding air. Ranges of various particle sizes may be estimated using aforementioned equations (Eqs. (22) to (25)).

Minimum pressure to obtain the desired velocity through the nozzle is

$$p_{an}=p_{\alpha}\left[1-v_{an}^{2}\frac{(k-1).g.k.R.T_{\alpha}}{2}\right]^{k(k-1)} \tag{26}$$

Where, p_{α} is the ambient pressure (bar), p_{an} is the pressure at the nozzle point, k is ratio of specific heat of gas at constant pressure and volume. Eq. (26) may be used for estimating the ranges of the particle injection and the pressure rating of the system.

Power Requirement (kW) for Rotary Feeder. Power requirement estimation for rotary air lock feeder can be carried out as follows [6,15]:

The force (N) required for moving the material against frictional resistance of rotor w.r.t. coal column in feeder hopper above feeder and crushing of coal particles cramped along the tip area of bucket/vane of the rotor and feeder housing:

$$F=F_f+F_c=h\rho_c g_c\frac{\Pi D_{ro}}{1000}\times\frac{b}{1000}\phi+C_s\frac{d_c}{1000}\times\frac{b}{1000}m \tag{27}$$

Where, ρ_c is density of coal particle (kg/m^3), D_{ro} and D_{ri} are the outside and inside diameters of the rotor (mm), b is the width of the bucket/vane (mm), h is the height of coal pile in feeder hopper above rotor/feeder, Φ is the ratio of maximum number of vanes m below feeder housing to that of total number, C_s is the crushing strength of the coal (N/m^2), d_c is the diameter of the maximum size of the coal (mm).

Therefore, power (kW) required to drive the feed $P_c=F\times\Pi D_{ro}/1000\times N/1000\times 60$

For this study, to estimate the actual power required for the motor, the power P is multiplied by 3 to allow for motor efficiency, rotor efficiency, bearing friction, resistance of any hard particle between bucket tips and feeder housing.

$$P_m=3P=3\times F\times\Pi\frac{D_{ro}}{1000}\times\frac{N}{1000\times 60} \tag{28}$$

Compressed Air for Injection. The necessary power required for supplying the air at the required pressure and flow rate can be estimated as follows:

$$Pc=\frac{1}{\eta_c}\cdot 2.78\times 10^{-4}\cdot\frac{k}{k-1}V_1p_1\left[\left(\frac{p_2}{p_1}\right)^{k-1/k}-1\right] \tag{29}$$

Where, η_c is the compressor efficiency about 60%, inlet pressure p_1 (kPa), V_1 is the volume flow rate (m^3/h), p_2 is discharge pressure and $p_2/p_1\leq 0.575$. Eqs. (21) to (30) are evaluated based on capacity of the plant. Using these equations, the acceptable power rating of the above mentioned components can be estimated. Plant wise coal and air distribution rate is presented in Table 4.

Estimation of Savings due to Proposed Kiln Seal Arrangement

Any ingress of air is absolutely unwanted for freshly reduced sponge iron at an elevated temperature. Therefore, the outlet end area of the rotary kiln and the inlet end area of the rotary cooler and the intermediate hood require well-designed air sealing systems. The mechanical sealing system as shown in Fig. 2 is the best arrangement in which the chance of any air ingress is almost negligible. Commercial plants are however, provided with the above mentioned type of mechanical sealing at both the end areas of the kiln and the cooler. Frictional resistance along the mechanical sealing surface contributes toward considerable energy consumption. The energy consumption (kW) between stationary parts and rotary parts of the mechanical seal at each area may be estimated by the following empirical equation [16]:

$$P_s=2\pi TN/60000 \tag{30}$$

Where T is the torque (N-m) and N is RPM of rotary part. The total torque at the gland area as rubbing area of the sealing system is as follows [16]:

$$T=\mu_g F_g D_g+\mu_s F_s D_s \tag{31}$$

Where, T is the torque (N-m), μ_g and μ_s are average coefficient of friction of gland and face/surface seal respectively, F_g and F_s are normal force (kN) at sealing surface at gland and shell, face/surface D_g and D_s are gland and shell diameters (mm).

This stringent sealing is not necessary at the inlet end of the kiln and the outlet end of the cooler. The proposed air jet type sealing is shown in Fig. 5. In this arrangement, the air enters through one end of the jet sealing device and is evenly delivered/discharged and distributed through uniform jets to form an air curtain/ring over the space between the stationary and rotary surfaces. This ring does not allow process gas to escape and makes the necessary sealing arrangement. Assuming the diameter of the

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rotary kiln to be 4.0 m, the clearance from the sealing area between the kiln and the hood varies from 1.5 mm to 2 mm. The air jet device/distributor should be placed suitably and the required jet speed for sealing is about 2–3 m/sec. The total air required for this purpose is of the order of 1000 to 1500 m^3/h at 10 mbar and system needs a blower of 5 to 6 kW rating. The system does not require maintenance or greasing between rubbing surfaces of the mechanical seal.

Use of Only High Pressure Atomized Water Supply in the Post Combustion Chamber

In order to make flue gas/exhaust emission free from suspended particulate material and toxic gases such as carbon monoxide, unburnt hydrocarbon, sulphur oxides etc., the flue gas leaving the kiln is initially mixed with fresh excess air and then burnt completely in the PCC by the application of a suitably short and stable flame. The temperature of the gas leaving the kiln therefore, increases in PCC, which needs to be brought down to a workable limit for down stream equipment required for the removal of dust particles and toxic components. Thus, water is sprayed to reduce the temperature of the flue gas in the PCC. Important parameters on which the cooling depend are the quantity of water sprayed, and evaporation rate and retention time of water droplets in free falling space within PCC. The evaporation rate of water particles increases with increase of its surface area i.e. with the decrease of the droplet size. Moreover, the retention time of the droplet in the gas stream also increases with the decrease in droplet size because of increased viscous drag force. Therefore, a highly atomized water spray is expected to give a better result.

The amount of water sprayed and the power required for circulation of the spraying water may be estimated as follows [17]:

$$M_f C_f (t_{fm} - t_{ff}) = M_w C_s (t_{ff} - t_{ws}) + M_w L + M_w C_w (t_{ws} - t_\alpha)$$

or

$$M_w = \frac{M_f C_f (t_{fm} - t_{ff})}{C_s (t_{ff} - t_{ws}) + L + C_w (t_{ws} - t_\alpha)} \tag{32}$$

Where, M_w is the rate of water spraying (kg/h), M_f is rate of flue generated (kg/h), C_f, C_w and C_s are specific heat of flue, water and supersaturated water vapor respectively (kJ/kg°C), L is latent heat of water (kJ/kg), t_{fm}, t_{ff}, t_{ws} and t_α are maximum flue temperature, final flue temperature, temperature of saturated vapor and ambient temperature (°C) respectively.

It is to be noted that the flue gas contains some unburned hydrocarbons/volatile when they leave. In case of a highly atomized system, water required for spraying will be 20 to 25% more. Further, high atomization requires the use of very high pressure water pumping (20–25 bar) system [18].

The power (kW) requirement is $p_w = \rho_g M_w H / 1000\eta$

Where H is the equivalent pressure head (m).

Some of the commercial plants are provided with coarser spray/droplet system as the system can be operated using lower pressure/head rating centrifugal pumps.

Discussion of Results

The entire heat transfer estimation carried out in this study is based on stabilized and optimized temperature data within the kiln of various sizes of sponge iron plants. Estimation of heating up cycle and cooling down period is not considered. Also, in the heat transfer analysis, the estimation of heat transfer from the bed surface to the interior of the bed by bulk mixing through tumbling action is not considered, as appropriate values of shape factor could not be ascertained. Moreover, Q_s, the heat transfer to the solid (Table 3) is found to be more that the Q_s, the estimated heat required for bed of material in Table 2. This difference is of the order of 10–15%. The mentioned difference certifies that higher amount of heat rate must be available for the required heating of the bed of the material. The mentioned procedure can also be utilized for the design estimation of various sizes rotary reactors for sponge iron making.

The maximum temperature of the smaller kiln (Table 1) is 1100°C where as that of larger sized kiln is about 1170°C. This is due to the fact that in the larger diameter kiln, the preheating time increases significantly as the ratio of surface area of charge bed across which heat transfer takes place and the volume charge decreases. Increasing heat transfer using higher temperature for sponge iron making may lead to inefficient operation and operational problems such as localized heating and liquid phase formation.

All the plants are provided with air tubes that deliver air at the desired rate from shell mounted blowers along the kiln axis towards either inlet or outlet and from inlet feed chute area.

Some commercial plants are provided with highly efficient submerged air injection facility in the preheating zone for effective burning/utilization of evolved volatile materials from coal within the bed of material and due to this, the kiln can be operated at lower temperature. However, submerged air injection requires specially designed on/off control valves in the air module circuit from a shell fan. These valves are operated through sets of strikers that are mounted on the stationary structure fixed on the kiln platform. Due to dusty and warm conditions, the valves often operate under dry lubrication. Moreover, due to kiln rolling back during power failure and kiln expansion, the striker and valves are subjected to malfunctioning. Therefore, considering the cost of vigil maintenance of valves and strikers, the radial injection that operates on 65–70% efficiency of the submerged air injection is proposed in this study.

As mentioned already, the reducing zone of the kiln is created and controlled by a pneumatic coal injector. Pneumatic injection uses kinetic energy/flow momentum from either more air at low velocity or less air at high velocity. As the reducing zone is adversely affected by excess air, use of less air is desirable. Further excess air used for coal injection requires excess equivalent coal that ultimately may result in the overheating of the kiln. The information available in literature on coal injection is meager and therefore, the present work is expected to be helpful for estimating optimized coal injection rate and power requirement.

The kiln inlet end as well as cooler outlet end areas may have an air jet type sealing arrangement as proposed in this study. Incorporation of such type of sealing in the aforementioned areas will not only reduce the power consumption of the kiln and cooler motor but also reduce the sizes of centralized grease lubricating system.

Some commercial plants are provided with coarser spray/droplet system for cooling of high temperature flue gas at PCC. Evaporating efficiency of coarser water droplet is very poor and requires very large quantity of water to reduce the temperature of flue gas to the desired level. About 80–85% of spray water particles are collected in the collecting chamber located at the bottom

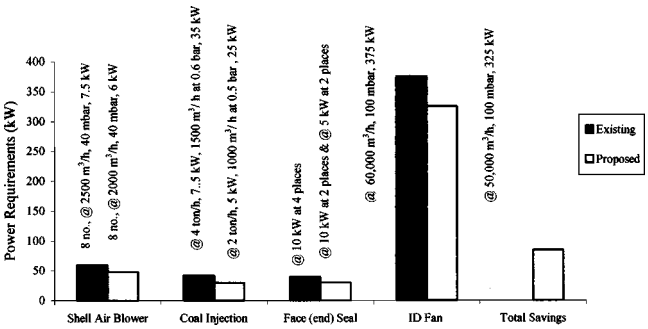


Fig. 6 Details of energy savings in 100 TPD DR sponge iron plant

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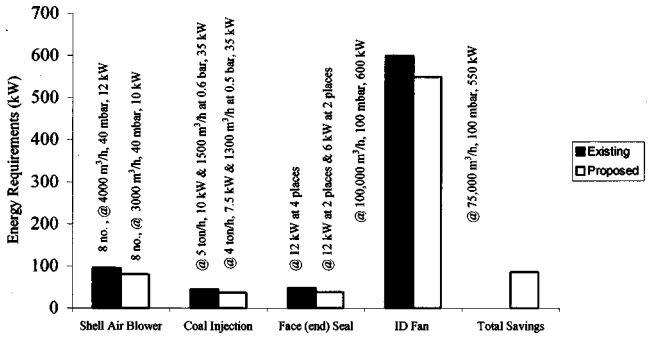


Fig. 7 Details of energy savings in 200 TPD DR sponge iron plant

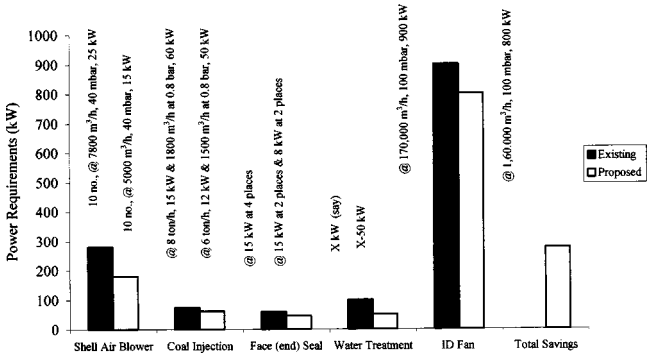


Fig. 8 Details of energy savings in 300 TPD DR sponge iron plant

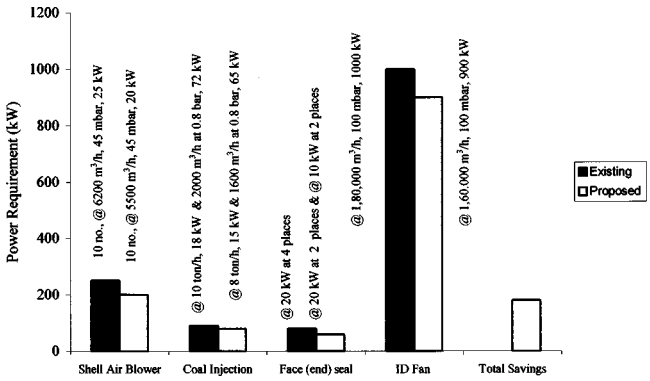


Fig. 9 Details of energy savings in 400 TPD DR sponge iron plant

of the PCC that subsequently requires treatment against contamination before being used in the re-circulating system. Droplet size spraying may be accomplished by adopting centrifugal pumps. But, high-pressure positive displacement pump is required to generate atomized spray. Therefore, if a plant is provided with high pressure atomizing, the system will reduce the size of water spraying unit as well as the size of the main water handling system. With the proposed reduction of the size of the shell blower as well as coal injection system, size of ID fan will also decrease significantly.

Figures 6–9 shows comprehensive summary of energy saving of various plants based on the analysis mentioned in the preceding sections.

Conclusions

A perusal of bar graph 6–9 reveals that for the 100 TPD plant, with 2.6 m kiln diameter and 40 m kiln length, the energy consumption due to the mechanical seal is 10 kW per sealing area. However, if the air ring sealing as proposed in this study is used, at the inlet end of the rotary kiln and outlet end of rotary cooler, the expected saving is 10 kW per area. The present energy consumption due to coal injection facility is 35 kW up to 4 tonne/h with 7.5 kW drive. However, 12.5 kW of savings is expected if the system adopted is optimized using air jet sealing around the kiln inlet and cooler outlet end with a low capacity blower. There is a potential for 1.5 kW savings for each shell blower due to optimization of shell air blower as found in this study. As there are 8 blowers, the expected energy saving is 12 kW. Because of reduction of shell blower capacity and coal injection facility, the total air handling capacity will reduce. Hence the ID fan capacity will also reduce significantly. The expected savings due to reduction of ID fan capacity is 50 kW. Total energy savings due to proposed modifications can be up to 85 kW for 100 TPD plants. Similarly, the possible energy savings for 300 TPD plants are worked out to be about 277 kW and for 400 TPD plants as 180 kW. Thus, the study presented in the paper finds considerable potential for energy savings in various capacity DR sponge iron plants and suggests mechanism to achieve it.

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